

CliniCause: An End-to-End Framework for Constructing Clinically Grounded Causal Analysis Datasets

Anonymous submission

Abstract

1 Introduction

Causal inference asks how outcomes would change under alternative actions, rather than only which outcomes can be predicted. This distinction is central to medicine and public policy: an accurate forecast may identify who is at risk, but it does not establish which intervention would alter that risk. Modern causal machine-learning methods support flexible adjustment and heterogeneous-effect estimation (Chernozhukov et al. 2018; Wager and Athey 2018; Shalit, Johansson, and Sontag 2017; Louizos et al. 2017), but evaluating them is unusually difficult. For each individual, the counterfactual outcome needed to score an estimated effect is unobserved.

Because observational data do not reveal counterfactual outcomes, they do not provide a direct answer key for evaluating causal estimators. They also leave the true treatment-assignment process and the extent of unmeasured confounding unknown. The safest experimental sandbox is therefore fully synthetic data: the evaluator specifies the covariates, treatment assignment, potential outcomes, confounding, overlap, and effect heterogeneity. Assumptions such as ignorability or positivity can be satisfied, violated, or gradually stressed under complete control, while the true causal effect remains known.

That control creates an important limitation. The evaluator determines not only the answer but also the difficulty of recovering it. A method may perform well because its assumptions align with the selected assignment model, response surface, confounding structure, or overlap regime. Semi-synthetic benchmarks move closer to observed data by retaining empirical covariates while simulating treatment assignments, potential outcomes, or both. RCT-derived and empirically anchored designs preserve still more of the observed system, but continue to obtain an answer key through researcher-controlled sampling, comparison, or generation mechanisms. Across this spectrum, greater control makes

accuracy measurable, while greater observational fidelity makes the causal target less knowable.

Several widely used causal-ML resources illustrate this spectrum of constructed evaluation. IHDP retains empirical covariates while simulating outcomes, whereas ACIC uses empirical covariates within diverse simulated treatment and outcome mechanisms; Dragonnet, for example, evaluates 101 retained datasets drawn from 63 ACIC 2018 data-generating settings (Shalit, Johansson, and Sontag 2017; Shi, Blei, and Veitch 2019). Influence-function validation similarly compares models across 77 semi-synthetic competition datasets sharing one empirical feature source but using distinct simulated assignments and outcomes (Alaa and van der Schaar 2019). More empirically grounded designs retain additional observed components: observational sampling from randomized trials preserves real trial variables but induces confounding through researcher-specified selection (Gentzel, Pruthi, and Jensen 2021); Jobs/LaLonde combines experimental treated subjects with external nonexperimental controls, making comparability depend on population-selection and harmonization choices; and Credence learns from an empirical distribution but generates data under user-specified effects and confounding (Parikh et al. 2022). These designs are increasingly sophisticated and provide valuable causal answer keys. Nevertheless, measurable accuracy still depends on an experimental anchor or a researcher-constructed causal, sampling, or comparison mechanism. They therefore cannot by themselves show how estimators behave when exposure prevalence, missingness, support, and measurement processes arise from routine practice rather than from the evaluation design.

In this paper, we propose a pivot in where researchers intervene when constructing causal-evaluation resources. Before defining this paradigm formally, consider a running ICU example in which the goal is to estimate how a shock state affects in-hospital mortality. The raw record does not contain a single, ready-made shock variable. Instead, it contains an irregular sequence of blood-pressure measurements, lactate tests, urine-output observations, and vasopressor administrations, followed by a source-observed mortality outcome.

A fully synthetic benchmark could generate the patients, assign shock according to a researcher-specified probability, and simulate mortality under shock and no shock. A semi-synthetic benchmark could retain observed patient covariates

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while simulating the shock assignment, mortality outcomes, or both. Either construction provides a known causal answer, but both the answer and the difficulty of recovering it depend on researcher-selected mechanisms.

We propose shifting this intervention from the data-generating process to the *representation layer*. In the running example, the evaluator preserves the shock-related measurements, clinical practice patterns, missingness, empirical support, and mortality recorded during routine care. The evaluator intervenes by specifying which measurements and temporal rules define the shock exposure, which patients enter the cohort, which variables serve as covariates and outcomes, and which causal relationships determine adjustment.

Crucially, these choices are knowledge-driven rather than learned from exposure–outcome associations in the evaluation data. Clinical experts, published evidence, or externally elicited domain knowledge specify, for example, that hypotension, elevated lactate, low urine output, and vasopressor use constitute evidence of shock; the dataset is not searched for the definition that produces the strongest mortality effect. Patient records are used to instantiate this prespecified representation. This separation reduces outcome leakage and post-hoc selection while making every threshold, temporal rule, cohort restriction, and graph edge an explicit knowledge claim that can be inspected, challenged, and replaced.

LLMs make this knowledge-driven construction increasingly scalable. Their broad clinical knowledge can support structured elicitation of candidate constructs and relationships (Singhal et al. 2023), while their reliability limitations require that every proposal remain auditable and subordinate to external evidence and expert review (Darvariu, Hailes, and Musolesi 2024). The LLM therefore serves as a design assistant rather than a source of causal ground truth: it helps make the proposed representation explicit but does not process patient records, generate exposures or outcomes, or select definitions according to their estimated effects.

The running example reflects the real setting studied in this paper. We instantiate the representation-centered paradigm through *CliniCause*, an evidence-tracked pipeline applied to longitudinal ICU records from MIMIC-III and PhysioNet 2012.

CliniCause uses ChatGPT through its desktop interface as a knowledge-elicitation assistant at the representation stage. ChatGPT was given the names of the columns available in the ICU time series and the target outcome, but not the patient-level records, variable distributions, exposure prevalences, causal estimates, or access to the environment containing MIMIC-III and PhysioNet. Drawing on clinical literature available through its pretrained knowledge and online search, it was prompted to propose clinically meaningful categorical proxy exposures, the measurement cutoffs and temporal rules needed to instantiate them, and a DAG relating the proposed constructs, observed covariates, and mortality. Every proposed proxy, cutoff, and DAG edge had to be accompanied by a clinical rationale and a supporting literature source, making each knowledge claim traceable and open to verification.

Kobi — we need a separate supplementary PDF documenting the clinical evidence used to construct the CliniCause

representation. Please prepare a LaTeX file containing two tables:

1. Proxy-definition table: one row for every proxy exposure in both MIMIC-III and PhysioNet. Include the proxy name, source columns, exact cutoff(s), temporal/aggregation rule, clinical interpretation, and the paper(s) that ChatGPT supplied as support. Please include a short paraphrase of what each source supports and its DOI/URL.

2. DAG-edge table: one row for every edge in each dataset-specific graph. Include the parent node, child node, clinical rationale, supporting paper(s) supplied by ChatGPT, and whether you independently verified that the cited paper actually supports that relationship. .

The authors inspected the proposed rationales and translated the selected proxies and relationships into explicit rules and graph assumptions, yielding states such as renal dysfunction, hepatic dysfunction, shock, respiratory failure, cardiac strain, and sepsis-related burden. Because the LLM had neither patient-level inputs nor access to the data environment, the exposure definitions could not be optimized against observed MIMIC-III or PhysioNet outcomes during elicitation. After construction was complete, we conducted a separate literature-based corroboration of the resulting effect directions (Table 1); this post-construction audit was not used to select the proxies, tune their thresholds, or construct the DAG.

The resulting resources contain 26,845 MIMIC-III and 7,993 PhysioNet patient records, with nine and ten admitted proxy exposures, respectively. Rather than merely compressing the underlying time series, each resource provides an explicit causal-analysis interface connecting source-observed measurements and mortality to candidate exposures, covariates, cohort rules, graph assumptions, adjustment choices, and provenance.

By preserving the naturally occurring measurement, exposure, support, and outcome processes, CliniCause gives up the known answer provided by controlled benchmarks. Its credibility must therefore come from converging evidence rather than comparison with causal ground truth. We examine the resources along four complementary axes: whether the candidate exposures and graph assumptions are clinically plausible; whether the rule-derived constructs are recoverable from the underlying measurements; whether distinct causal estimators agree once the representation is fixed; and whether the estimated directions corroborate mortality relationships reported in independent clinical studies. Together, these checks assess whether CliniCause provides a convincing and useful observational causal testbed, without treating its proxies as literal interventions, its DAG as known to be correct, or its estimated effects as ground-truth causal answers.

We evaluate the resulting resources along each of the four proposed axes. Informal consultations with ICU physicians provided formative feedback on the clinical plausibility of the proxies and graph assumptions, while predictive experiments assessed whether the rule-derived proxies could be recovered from the underlying measurements. Across 19 pre-specified dataset–exposure comparisons, CausalForestDML and LinearDML agreed in direction in all 19, and CausalPFN

agreed in 18; PhysioNet shock was the sole exception. Finally, comparison with published mortality gradients supported the renal, hepatic, cardiac, inflammation/sepsis, and global-severity directions, while identifying shock and PhysioNet coagulation/hematologic dysfunction as clinically discordant cases requiring further review (Table 1). The physician consultations were informal rather than structured clinical adjudication, and the combined checks provide converging evidence rather than causal ground truth.

Our contributions are:

1. an evidence-tracked pipeline that converts irregular clinical records into explicit causal-analysis representations while preserving observed patient, support, and outcome processes;
2. two reusable resources derived from MIMIC-III and PhysioNet 2012, containing 19 clinically motivated proxy-exposure analyses with inspectable cohort, graph, adjustment, and provenance assumptions; and
3. an evaluation framework for observational causal resources based on construct plausibility, proxy recoverability, cross-estimator agreement, and clinical-literature corroboration, including explicit identification of discordant results.

2 Related Work

3 Constructing the CliniCause Datasets

CliniCause constructs two reusable, validated causal-analysis datasets from MIMIC-III and PhysioNet 2012. Each connects irregular observations to proxy exposures, in-hospital mortality, observed covariates, and inspectable graph metadata under a shared interface that retains source-specific semantics (Figure 1; Table 2).

3.1 Source Cohorts and Analysis Units

The resources originate from MIMIC-III and the PhysioNet/Computing in Cardiology Challenge 2012 collection (Johnson et al. 2016; Silva et al. 2012). An analysis unit is a source time-series record or ICU stay, not necessarily a unique person. Source identifiers map to a canonical `ts_id`; irregular events retain time, variable, and value, while a record table supplies mortality and available baseline covariates. Linked long-event and record-level views support temporal modeling and causal analysis without flattening irregular events into the estimator table prematurely. Split-aware predictive artifacts and the complete admitted causal cohort remain distinct and are aligned by validation contracts rather than treated as interchangeable files.

The admitted populations contain 26,845 MIMIC-III and 7,993 PhysioNet records. Their covariates, measurement processes, proxy ontologies, and graphs remain dataset specific, so records and estimates are never pooled and similarly named constructs are not assumed equivalent. The common interface standardizes analytical roles—identifier, exposure, outcome, covariate, adjustment, and provenance—rather than forcing a common clinical vocabulary onto the two sources. Portability can therefore be evaluated without

erasing source-specific measurement and construct boundaries.

3.2 Proxy-State Design and Deterministic Construction

Before patient-level execution, a structured LLM protocol proposed proxy ontologies, binary-rule families, missingness considerations, and dataset-specific graphs. Project-selected proposals were encoded in deterministic tagger and graph source. That source, not prompt text, governs runtime; the LLM neither processes patient records nor assigns exposures or estimates effects. This boundary makes proposals inspectable while ensuring that patient-level execution depends on versioned deterministic artifacts.

Dataset-specific rules map measurements and care-process indicators to binary proxy fields while preserving source differences (e.g., combined MIMIC-III hepatic/coagulation evidence versus separate PhysioNet constructs). Missing numeric measurements provide no positive rule evidence, making availability part of construct provenance. These are transparent analytical proxies, not chart-adjudicated diagnoses. Their explicit columns provide a replaceable interface: reviewers can revise a rule without silently changing the outcome, covariates, or cohort contract. Missing measurements cannot create positive rule evidence by being silently interpreted as normal; the observed input envelope remains part of each proxy’s meaning.

3.3 Predictive Annotations from Irregular Time Series

Each predictive task estimates its dataset’s rule-derived proxy vector. STraTS represents sparse observation times, values, and variables; GRU and TCN process temporal grids; and GRU-D models decay from missingness and time gaps (Tipirneni and Reddy 2022; Cho et al. 2014; Bai, Kolter, and Koltun 2018; Che et al. 2018). Static covariates join each temporal representation. Value, observation-mask, and elapsed-time channels preserve different aspects of irregular sampling across architectures. Targets are proxy states, not mortality.

Normalized exports pair target probabilities with binary predictions at the implemented 0.5 threshold. Archived causal runs combine one rule-derived source and four aligned model sources by deterministic majority vote. This aggregate is an algorithmic proxy-exposure layer, not a learned label model or expert consensus. Predictive metrics characterize recovery of rule targets, not their clinical validity. Probability and binary fields remain paired in normalized exports, while aggregation consumes only aligned binary voters; prediction evaluation therefore remains distinct from exposure assembly. Complete voter sets are required so that a missing model export cannot silently change the aggregation denominator or admitted cohort.

3.4 Causal-Analysis Schema and DAG Metadata

Estimator-ready rows align canonical identifiers, aggregate proxy fields, in-hospital mortality, and source-specific covariates. Linked metadata records dataset, sampling condition, graph, run, and artifact provenance. The analysis admits

Table 1: Clinical-literature corroboration of selected original-cohort CATE estimates. Values are mean estimated changes in in-hospital mortality probability, in percentage points, for CausalForestDML (F), LinearDML (L), and CausalPFN (P). Literature findings provide directional and clinical-plausibility comparators, not ground-truth causal effects or direct calibration targets.

Proxy variable	MIMIC-III CATE (pp)	PhysioNet CATE (pp)	Clinical literature comparator and interpretation
Renal dysfunction / AKI	F: +9.1; L: +8.9; P: +8.7	F: +12.0; L: +9.1; P: +11.1	KDIGO defines AKI using creatinine and urine-output changes. AKI-EPI reports a stage-dependent mortality gradient, reaching an adjusted mortality OR of 6.884 for stage 3; a septic-ICU cohort attributes approximately 16.8 percentage points of 30-day mortality to stage 3 AKI. The positive CATE direction and rough magnitude are clinically plausible, conditional on the proxy capturing moderate or severe AKI.
Hepatic dysfunction / bilirubin	F: +9.8; L: +8.3; P: +9.7 ^a	F: +9.1; L: +6.0; P: +10.7	ICU studies report higher mortality with elevated bilirubin: 30% versus 21% for bilirubin above 1 mg/dL, and 31.9% versus 17.0% for bilirubin at least 2 mg/dL in MIMIC-III. The positive +6 to +11 pp estimates are directionally and approximately quantitatively consistent.
Cardiac strain / injury	F: +22.0; L: +18.8; P: +25.9	F: +11.2; L: +12.2; P: +11.9	A review of ICU troponin studies reports an adjusted mortality OR of 2.5 for elevated troponin; a sepsis meta-analysis reports mortality HR 1.57. The positive direction is supported, although the literature does not establish that cardiac strain should be the highest-ranked isolated causal pathway, particularly in MIMIC-III.
Inflammation / sepsis burden	F: +16.1; L: +16.1; P: +14.9	F: +7.2; L: +7.1; P: +6.7	Sepsis-3 associates infection-related organ dysfunction with in-hospital mortality exceeding 10%. The positive estimates are clinically plausible, but the large MIMIC-III estimate may indicate that the proxy captures organ dysfunction or measurement intensity in addition to inflammation (Singer et al. 2016).
Global severity	F: +6.2; L: +8.0; P: +7.3	F: +10.8; L: +10.7; P: +10.5	SOFA was developed to describe organ dysfunction and failure, and increases in SOFA are strongly associated with mortality. The consistently positive estimates, particularly the approximately +11 pp PhysioNet estimates, are directionally plausible (Vincent et al. 1996; Singer et al. 2016).
Shock	F: +2.1; L: +2.1; P: +3.2	F: -1.4; L: -2.7; P: +0.4	Sepsis-3 septic shock requires vasopressors and lactate above 2 mmol/L after resuscitation and is associated with hospital mortality above 40%. The small MIMIC-III estimates and negative or near-zero PhysioNet estimates are <i>not</i> clinically aligned with a total shock effect. This is an explicit red flag for proxy construction, temporal definition, mediator adjustment, or available treatment information (Singer et al. 2016).
Coagulation / hematologic dysfunction	- ^b	F: +1.1; L: +0.4; P: +2.5	PLOT-ICU reports 90-day mortality of 34.6% among patients with thrombocytopenia versus 19.3% without it, rising to approximately 51–52% in severe cases. The near-zero PhysioNet estimates are substantially weaker than this clinical gradient and constitute a second explicit proxy-validation red flag.

^a The MIMIC-III variable combines hepatic and coagulation dysfunction and is therefore not directly equivalent to the PhysioNet hepatic proxy. ^b MIMIC-III has no separate coagulation/hematologic proxy; its mixed hepatic/coagulation variable cannot independently corroborate this clinical construct. F = CausalForestDML; L = LinearDML; P = CausalPFN; pp = percentage points. CausalPFN is exploratory and lacks the downstream diagnostics available for the two DML estimators.

nine non-chronic MIMIC-III and ten non-chronic PhysioNet exposures; chronic-burden fields remain baseline constructs rather than analyzed exposures. Exposure, outcome, adjustment, and provenance roles are explicit even when their fields reside across linked tables and metadata.

For each exposure, a source-coded graph maps project assumptions to observed columns and an exposure-specific adjustment set. The graphs are neither learned nor clinically validated ground truth, but encoding them makes adjustment choices inspectable and replaceable alongside proxies and estimators without rebuilding source-data integration. Alternative graphs can consequently be compared against the same aligned resource instead of being conflated with a new preprocessing population. Graph or estimator changes thus remain visible analytical choices rather than hidden changes to the data resource.

3.5 Dataset Validation and Provenance

Structural validation checks files, column order, identifiers, values, and output completeness. Cohort contracts require identifier preservation and exact equality at handoffs, rejecting duplicates, conflicts, missing or extra records, and silent shrinkage. Unexpected columns, invalid voter/probability fields, binary inconsistencies, and incomplete artifact sets fail explicitly. Together, these checks protect one analytical population from deterministic labels through predictive annotations and estimator-ready assembly.

Validation also spans artifact identity and execution boundaries. Prediction, checkpoint, and voter sidecars bind model and input roles; run receipts distinguish planned from completed stages; upstream fingerprints prevent reuse after source mutation; and dataset-isolated paths prevent cross-source artifact substitution. A handoff is validated only when its required evidence is present, rather than inferred from a

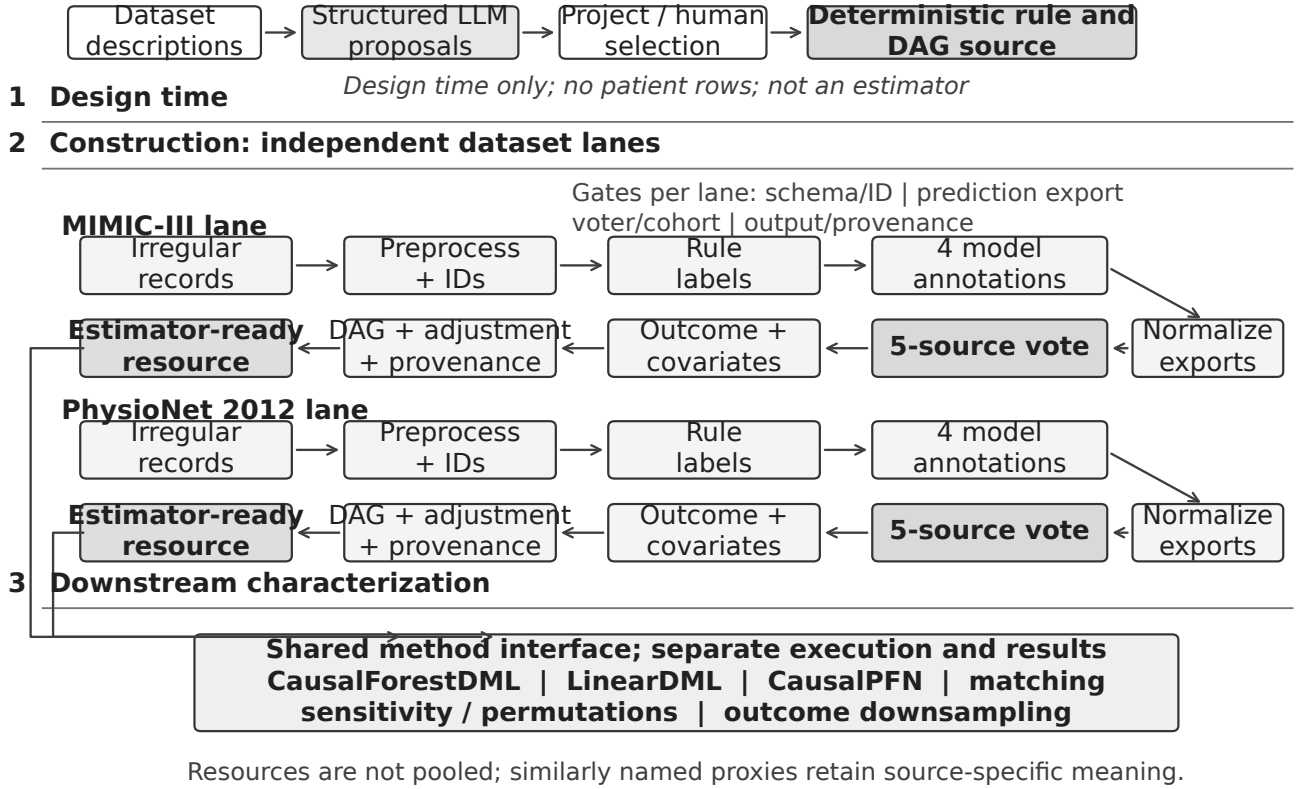


Figure 1: CliniCause construction. Separate dataset lanes combine deterministic proxy rules and four normalized model annotations, then package five-source aggregates with outcomes, covariates, graph metadata, and provenance. Validation gates protect each handoff. The LLM proposes candidates only at design time; selected deterministic source governs runtime.

file name or partial directory.

Artifact validation binds outputs to supported metadata, hashes, schemas, cohorts, configurations, and seeds. Dataset-isolated run paths, manifests, receipts, and sidecars record roles and outputs; stale or mutation-mismatched artifacts fail reuse checks. Deterministic transformations and derived seeds avoid dependence on ambient order or process state. These contracts concern the current resource interface and do not recover historical run provenance by themselves.

The checked archive establishes valid export schemas, complete primary aggregates, manifests, checksums, and archived analysis-stage records, but exact producing revisions, configurations, and checkpoint-to-export links remain incomplete. The repaired current code contains stricter source/test contracts; it is not claimed as the historical producer, nor do we claim current test execution. Keeping these evidence layers separate avoids turning checked output integrity into an unsupported exact-rerun claim.

Both resources supported predictive characterization, three effect estimators, graph-guided adjustment, matching, and robustness diagnostics. Validation thus establishes structural integrity, cohort consistency, provenance, and analytical reuse; clinical construct validity and causal identification remain separate questions. This scoped definition per-

mits strong resource validation without promoting proxy states to diagnoses or adjusted contrasts to proven effects. For current-code reproduction, one top-level launcher exposes dedicated dataset-construction and complete-pipeline modes. The supplement documents the Python 3.10 environment, single dependency installation, authorized data paths, validation, manifests, receipts, and provenance needed to reconstruct the derived resources; missing producing lineage still precludes a verified historical rerun, and an approved anonymous repository location remains gated.

4 Empirical Evaluation

The evaluation asks three linked questions. First, can irregular-time-series models recover the constructed rule targets, and does their behavior differ by dataset? Second, are the principal exposure–outcome directions stable across distinct estimator forms? Third, do matching and population perturbations expose support or sign sensitivity? These questions characterize the resources and their analytical interfaces; they do not test a clinical intervention.

4.1 Prediction Tasks and Metrics

For each dataset, STraTS, GRU, GRU-D, and TCN predict its deterministic proxy vector. We compare one selected

Dataset	Records	Exposures	Outcome	Predictive annotations	Causal-analysis estimators	
MIMIC-III	26,845	9	In-hospital mortality	1 rule source + 4 models	CF-DML, CausalPFN	LinearDML,
PhysioNet 2012	7,993	10	In-hospital mortality	1 rule source + 4 models	CF-DML, CausalPFN	LinearDML,

Table 2: Primary original-sampling resources. Counts are analysis records and admitted non-chronic proxy exposures, not raw cohorts; CF-DML denotes CausalForestDML.

archived held-out test output per model–dataset combination, without pooling targets or scores. Because repeated-run uncertainty is unavailable, these point metrics characterize prediction of rule-derived constructs, not statistical superiority or clinical validity. Target sets follow the source-specific schemas, so the evaluation asks whether each constructed proxy vector can be recovered from its own irregular ICU representation. The comparison includes all eight archived model–dataset outputs selected for the main analysis rather than choosing models by a single favorable metric.

We report sample-mean multi-label binary cross-entropy loss and macro AUROC, AUPRC, and minRP. Per target, minRP maximizes the smaller of precision and recall over thresholds. The three discrimination scores are averaged after excluding single-class targets; loss follows the archived sample-mean evaluator output. No confidence interval or significance test is inferred from the selected runs.

4.2 Effect Estimators and Cross-Estimator Evaluation

Each admitted aggregate proxy is analyzed separately as a binary exposure, with in-hospital mortality as outcome and observed adjustment variables selected by the project graph. Every prespecified non-chronic exposure is retained. The original populations are primary, and datasets are analyzed without pooling. This complete-set policy prevents favorable post-hoc exposure selection. The graphs encode the project’s observed-adjustment assumptions; they do not remove requirements for consistency, overlap, well-defined interventions, or absence of unmeasured confounding.

CausalForestDML is the primary nonlinear estimator, combining DML adjustment with a causal-forest final stage (Chernozhukov et al. 2018; Wager and Athey 2018; Athey, Tibshirani, and Wager 2019; Oprescu et al. 2019) to obtain record-level fitted conditional effects. LinearDML retains the DML nuisance framework with a linear final effect model, isolating a change in final-stage form. CausalPFN (Meresht et al. 2025) provides a distinct complementary paradigm over the same prespecified comparisons; its archived estimator-specific sensitivity and permutation coverage is less complete than for DML.

We summarize fitted conditional estimates by their *mean model-estimated CATE over the analyzed sample*, not as a risk ratio, odds ratio, or unqualified population ATE. We compare its sign between the DML estimators and across all three estimators, retaining disagreements. Directional agreement neither requires equal magnitudes nor proves causal identification; it tests whether the principal sign pattern de-

pends on one estimator family. All interpretations are therefore observational and proxy based, and none is a treatment recommendation. Magnitude remains visible in Figure 2 so sign concordance does not collapse substantively different fitted summaries into equivalence.

4.3 Matching and Robustness Diagnostics

Matching is a descriptive support check: binary adjustment variables and median-binarized numeric variables enter greedy one-to-one Hamming matching. The procedure begins with exact binary matches and increases allowed distance within its configured pair-count and match-rate criteria. We report a *descriptive matched-pair outcome difference* with availability, distance, and support warnings; no usable representation produces an explicit failure, not a zero effect or an imputed estimate.

Archived DML sensitivity records retain native, recomputed/reconstructed, partial, failed, and unavailable provenance classes. Treatment and outcome permutations are disruption checks rather than formal randomization tests or identification proofs. Equivalent stages were not archived for CausalPFN, which marks diagnostic-coverage asymmetry rather than exclusion from triangulation. For DML, recorded sensitivity provenance is preserved because estimator-native, later reconstructed, partial, and failed evidence are not interchangeable forms of validation.

Outcome downsampling retains mortality-positive records and samples negatives, changing both population and prevalence. It is a robustness perturbation, not a co-primary analysis, and its magnitudes are not interchangeable with original-population estimates. Applying the shared sequence separately to both resources tests workflow portability while preserving dataset-specific definitions, cohorts, and causal assumptions. The matching, disruption, and downsampling analyses are supporting diagnostics; no single check upgrades adjusted observational contrasts into identified intervention effects.

5 Results

The resources comprise 26,845 MIMIC-III records with nine exposures and 7,993 PhysioNet records with ten. All analyses remain dataset specific and unpooled.

5.1 Predictive Characterization

STraTS led all four archived MIMIC-III point metrics, whereas GRU-D led all four PhysioNet metrics (Table 3). This dataset-dependent leadership does not imply statistical superiority. STraTS reached 0.905 AUROC and 0.869

AUPRC on MIMIC-III; GRU-D reached 0.918 and 0.905 on PhysioNet. Each also led loss and minRP within its dataset, so leadership was consistent across the four archived metrics against rule-derived targets. The change in leader reinforces the resource design choice to retain source-specific measurement and missingness semantics rather than select one temporal representation globally.

5.2 Effect Patterns and Estimator Agreement

All nine MIMIC-III CausalForestDML mean summaries were positive, as were nine of ten PhysioNet summaries. MIMIC-III was led by cardiac strain (0.220) and inflammation/sepsis (0.161); PhysioNet was led by renal dysfunction (0.120), cardiac injury/strain (0.112), and global severity (0.108), while shock was negative (-0.014). These are mean model-estimated CATE over the analyzed samples under the implemented proxy and adjustment design; magnitudes and construct orderings remained dataset specific (Figure 2).

CausalForestDML and LinearDML agreed in direction for all 19 prespecified comparisons. CausalPFN joined their direction in 18/19, providing triangulation from a distinct modeling paradigm while retaining its smaller archived diagnostic envelope. Complete DML concordance shows that the sign pattern was not unique to a linear or forest final stage, while CausalPFN's broad agreement extends the observation beyond that shared DML framework. Neither result implies equal magnitudes or identifies a causal effect.

PhysioNet shock was the sole exception: CausalForestDML was -0.014 , LinearDML -0.027 , and CausalPFN 0.004; its descriptive matched-pair difference was 0.010. The disagreement motivates targeted analysis of the proxy, support, adjustment assumptions, and estimator behavior rather than estimator voting. Retaining this near-zero exception is important because a direction-only summary would otherwise conceal exactly where model choice changes the conclusion.

5.3 Robustness and Cross-Dataset Findings

Matching succeeded for 15/19 comparisons (eight MIMIC-III and seven PhysioNet); four failures lacked usable binary columns and are not zero effects. Of the 15 summaries, 14 agreed with the primary direction; PhysioNet shock did not. The availability split also distinguishes empirical support limitations from estimator disagreement on completed analyses. Matching agreement is descriptive corroboration only: it neither repairs failed overlap nor supplies an independent identified causal effect.

Direction persisted in 55/57 original-versus-downsampled comparisons. The two changes were PhysioNet LinearDML coagulation/hematologic dysfunction and PhysioNet CausalPFN shock; magnitudes do not transport to the original population. DML sensitivity and permutation evidence has nonuniform provenance and coverage, with no equivalent archived CausalPFN stages; these diagnostics remain supporting evidence rather than independent identification tests. Together, differing predictive leaders and effect patterns alongside broad estimator agreement show workflow portability, not pooled clinical replication.

Dataset	Model	Test loss↓	Test AUROC↑	Test AUPRC↑	Test minRP↑
MIMIC-III	STraTS	0.348	0.905	0.869	0.795
	GRU	0.386	0.881	0.840	0.770
	GRU-D	0.404	0.884	0.841	0.773
	TCN	0.414	0.867	0.823	0.758
PhysioNet 2012	STraTS	0.341	0.915	0.877	0.818
	GRU	0.397	0.915	0.897	0.831
	GRU-D	0.331	0.918	0.905	0.835
	TCN	0.477	0.899	0.875	0.811

Table 3: Selected archived held-out point metrics against rule-derived proxy targets. Bold marks each dataset’s best value; no significance is implied.

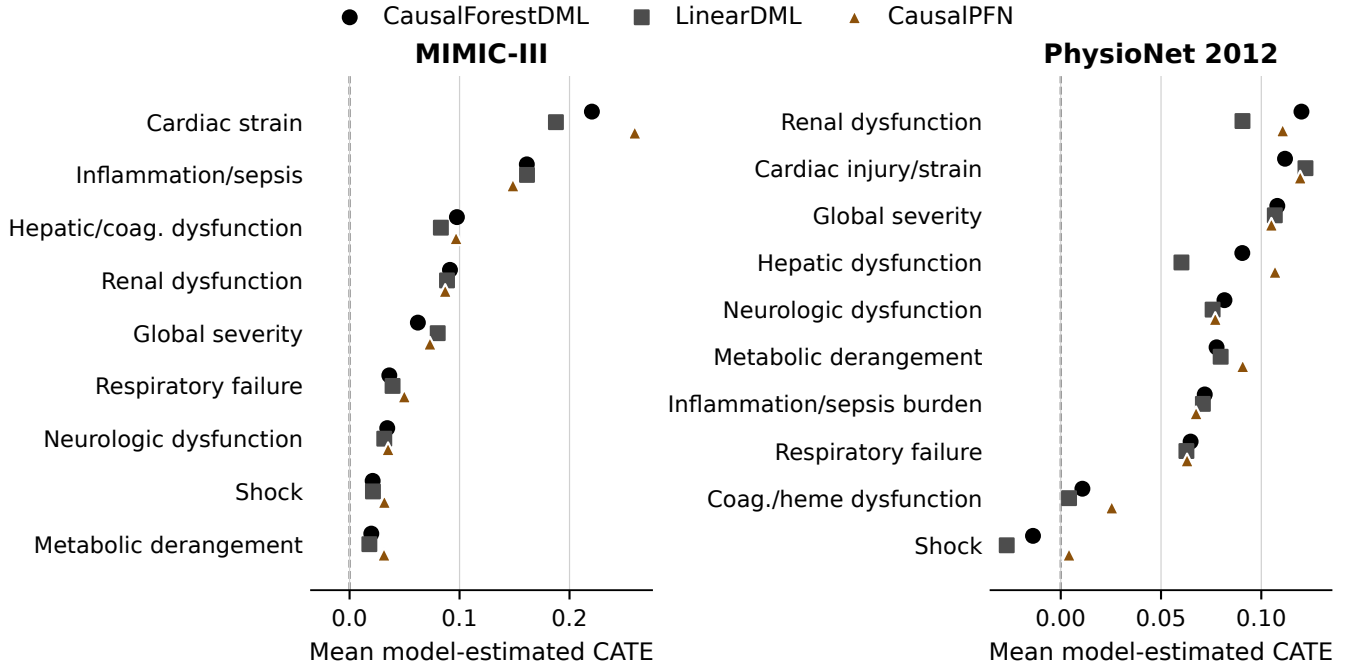


Figure 2: All 19 original-cohort comparisons, ordered by CausalForestDML. Markers are mean model-estimated CATE over each analyzed sample. DML directions agree in 19/19 and all three in 18/19; PhysioNet shock is the exception.

6 Discussion and Limitations

6.1 Resource Utility and Cross-Dataset Portability

CliniCause contributes reusable resources rather than only effect tables. Explicit exposure, outcome, covariate, graph, cohort, and provenance interfaces support comparisons of estimators and temporal representations without repeating every integration stage. Deterministic definitions and lineage also make proxy, graph, adjustment, cohort, and support alternatives directly auditable. A study can replace one layer while retaining the remaining analytical contract, making the present estimates starting points for controlled follow-up rather than fixed endpoints. Concretely, the same tables can support estimator benchmarking; the event-to-proxy interface can test new temporal representations; and preserved construct and graph metadata can support clinician-led proxy

or adjustment revision. Each use keeps cohort identity visible, helping distinguish a method change from an unnoticed population change. Because resource validation is separated from clinical and causal validation, follow-up studies can strengthen one evidence layer without discarding the audited interfaces already established in the others.

The validation contracts are scientifically consequential rather than only software safeguards. Exact cohort equality prevents a join from silently changing the population summarized by an estimator. Schema and voter checks prevent missing annotations from redefining an exposure. Graph and adjustment metadata expose assumptions that would otherwise be implicit, while hashes and receipts distinguish a known artifact from a similarly named replacement. Together these controls make differences between analyses attributable to declared choices instead of undetected pipeline drift. They do not establish that a proxy or graph is correct, but they make

those scientific questions answerable on a stable resource.

The same sequence operated across heterogeneous ICU sources although predictive leadership, magnitudes, and rankings differed. Portability therefore resides in the workflow and interface, not construct equivalence: variables, measurement processes, ontologies, graphs, and analyses remain dataset specific and unpooled. This separation exposes contextual sensitivity instead of hiding it behind a pooled replication claim. It also lets future work distinguish success of a common interface from equivalence of source semantics. Operation on two sources demonstrates workflow portability, not external clinical generalization of any proxy or effect estimate. The source differences also illustrate why a common interface is preferable to a forced common ontology. MIMIC-III and PhysioNet vary in available variables, sampling processes, and whether hepatic and coagulation evidence form one proxy or two. Preserving those differences allows researchers to compare methods while tracing disagreement to representation, construct, adjustment, support, or estimator choices. A pooled table would make that attribution harder and could give similarly named columns a false clinical equivalence.

6.2 Estimator Triangulation and CausalPFN

The 19/19 DML and 18/19 all-estimator sign agreements show that the prevailing pattern extends across final-stage forms and to CausalPFN’s distinct paradigm. PhysioNet shock usefully exposes model and support sensitivity for targeted study. Directional agreement does not establish equal magnitudes, uncertainty, or causal identification; CausalPFN’s smaller archived diagnostic envelope also motivates estimator-specific follow-up. Broad concordance plus one visible exception is more informative than an aggregate vote: it supports further CausalPFN evaluation while identifying a concrete case for proxy, adjustment, overlap, and estimator analysis. For future estimator studies, the complete 19-comparison matrix supplies both a broad concordance target and a prespecified stress case. Repeated runs, estimator-specific uncertainty, and harmonized diagnostics should determine whether the shock disagreement is sampling variation, weak support, construct sensitivity, or a genuine modeling difference.

6.3 LLM-Assisted Design and Future Research

The LLM layer is a design aid for inspectable proxy, missingness, and graph proposals; project selection and deterministic source retain authority. Future work can use the resources to test clinician-reviewed proxies, expert/LLM or design-layer alternatives, graphs, repeated-run uncertainty, richer CausalPFN diagnostics, and external datasets. Because proposals remain separate from patient execution, such comparisons need not grant the LLM runtime clinical authority. Prompt sensitivity and proposal-to-source traceability should also be evaluated so that any design benefit is distinguishable from downstream human selection and deterministic implementation.

6.4 Limitations

Proxy states are analytical constructs, not chart-adjudicated phenotypes, but their definitions and lineage permit clinician-led replacement and validation. Project graphs do not prove intervention validity; observational estimates remain vulnerable to unmeasured confounding, identification, and support failures and are not treatment recommendations. Alternative graphs, adjustment sets, and stronger support analyses are necessary before stronger interpretation. Repeated-run uncertainty and diagnostic coverage are incomplete, especially for CausalPFN. Exact producing revisions, configurations, and checkpoint-to-export lineage also remain incomplete; current runtime and anonymous release are separately gated. Accordingly, archived point-metric leaders are not statistical rankings, the agreement counts are not uncertainty estimates, and the present artifacts do not support an exact clean-checkout reproduction claim. Restricted source-data access, absent external clinical validation, and no isolated LLM ablation further limit generalization. The resource interfaces make these boundaries explicit and testable rather than burying them in undocumented pipeline choices. Replaceability facilitates those studies but is not itself evidence that a replacement, proxy, or graph is clinically valid.

7 Conclusion

CliniCause constructs reusable, estimator-ready resources from MIMIC-III and PhysioNet 2012 with validated structural, cohort, provenance, and analytical interfaces. Broad DML/CausalPFN directional agreement demonstrates bounded estimator triangulation: the DML estimators agreed in 19/19 directions and CausalPFN joined them in 18/19, with PhysioNet shock retaining the informative exception. Archived predictive leadership also differed by dataset, reinforcing the need to preserve source semantics. These transparent resources support tests of proxy definitions, temporal representations, graphs, cohorts, and estimators while preserving the observational and proxy-based boundary of the findings. This foundation is intended for transparent comparison and revision, not patient-level deployment or treatment selection. Its value lies in making the next proxy, graph, representation, or estimator comparison easier to specify, audit, and interpret.

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